Step 2a - H10 - Higher capacity mineral drying

Market scan

Market size

Same underlying pool as H9: mineral drying equipment+services is order-of-magnitude ~$1-3B/yr globally inside a ~$5-10B industrial dryer market, with minerals/ores ~24% of rotary dryer demand. The capacity-booster angle targets the installed base rather than new lines: dryers run 20-40 years, replacement/upgrade is a named demand driver in market reports, and OEM aftermarket/service is ~30-40% of dryer-OEM revenue (~). A new large rotary line costs ~$8-15M installed with 12-18-month lead times, so a retrofit that adds ~20-30% throughput for low single-digit $M addresses a real capex-avoidance budget — adjacent precedents exist (microwave boost ovens +25% throughput in food; IR booster sections debottlenecking paper machines; parallel fluid-bed additions at a 48->100 ktpa magnesium plant).

Region	Size ($B)	Region	Size ($B)
Europe	~$0.3-0.7B (aging installed base + ETS pressure; retrofit-friendly, capex-constrained mid-size producers)	Asia	~$1-2B (largest market but dominated by new-build/greenfield; retrofit slice smaller share)
America	~$0.5-1.2B (large aging fleet; debottlenecking culture strong in NA minerals/mining)	Africa	~$0.1-0.3B (mine expansions; often cheaper to overbuild new than retrofit)

Target customers

Operators whose drying step limits line throughput — typically because feed moisture rose (wetter ore, seasonal stockpiles), demand grew, or the dryer aged. Same base as H9: ~750 IMA-member processing plants in Europe, majors like Omya (~160 plants) and Imerys (~100+ sites) centralizing engineering decisions, plus hundreds of ore concentrators running single-line dryers where +10-20% capacity has high marginal value against $8-15M for a parallel line. Reachable through dryer-OEM aftermarket channels, plant engineering managers, and industry consultants; buying trigger is a capacity project already being scoped, so timing windows are episodic but the value case (avoided capex + deferred permitting) is concrete.

Customer size	Approx. number	Market share (in %)
Large	~30-50 producer groups worldwide, ~500-800 drying plants (~)	~50-60% of dried mineral tonnage (~)
Medium	~300-500 regional producers, ~1,000-2,000 plants (~)	~30% of tonnage (~)
Small	~1,000-3,000 single-site quarry/processors (~)	~10-15% of tonnage (~)

Players offering a solution today

Conventional debottlenecking is served by the incumbent dryer OEMs' aftermarket arms (Metso, FLSmidth, Andritz, ThyssenKrupp, GEA) selling upgrades, heat-recovery retrofits (~18-25% fuel savings reported in EU cement/mineral retrofits) and parallel units, plus mid-tier OEMs doing add-on dryers.

Booster/inline technology is a niche: microwave boost ovens proven in food (Ferrite/IMS MIP-9, +25% throughput), IR booster sections proven in paper/board (Ircon Solaronics), but no established microwave-booster player for bulk minerals — whitespace similar to H9.

Player	Type of solution	Market share (in %)
Large OEM aftermarket (Metso, FLSmidth, Andritz, ThyssenKrupp, GEA)	dryer upgrades, indirect-heat & heat-recovery retrofits, parallel lines, service contracts	~35-50% of new dryer revenue; aftermarket ~30-40% of OEM sales (~)
Medium (Allgaier, Comessa, Schenck, Carrier Vibrating, FEECO, Schwing Bioset, Hazemag, Louisville)	add-on rotary/FBD dryers, refurbishment, regional debottlenecking projects	~30-40% (~)
Small booster/dielectric niche (Ferrite/IMS, SAIREM, Pueschner, Cellencor; IR specialists e.g. Ircon Solaronics)	microwave/IR booster and pre-dryer modules 6-300 kW; proven +25-30% throughput in food/paper, mineral refs scarce	<5-10% today, emerging (~)

Regulation

Regulation	Jurisdiction	What it means for us
EU ETS (Directive 2003/87/EC)	EU	Debottlenecking by adding gas-fired capacity increases ETS exposure (~EUR60-90/tCO2); an electric booster adds throughput with zero on-site combustion.
MCPD 2015/2193 / IED permit review	EU	Uprating burners or adding combustion capacity can trigger permit modification and tighter emission limits; an electric booster avoids reopening combustion permitting.
US EPA NSPS/NESHAP air permitting	US	Increased firing rates on existing dryers can trigger New Source Review/BACT analysis; non-combustion boosters sidestep it.
Energy Efficiency Directive (EU) 2023/1791	EU	Mandatory energy audits push operators to exploit low-capex efficiency/capacity measures — a booster that raises output per GJ is a natural audit action.
National decarbonization funding (DE EEW, EU Innovation Fund, US IRA 48C/DOE)	EU, DE, US	Retrofit projects combining capacity gain with electrification qualify for process-heat decarbonization subsidies, improving booster economics.

Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input

Name	Devin (agent draft)	Devin (agent draft)	Date	2026-10-30	2026-10-30	Name			Date	
Decision	Decision	Go to next step	Go to next step	Go to next step	Rework	Rework	Rework	Drop the hypothesis	Drop the hypothesis	Drop the hypothesis